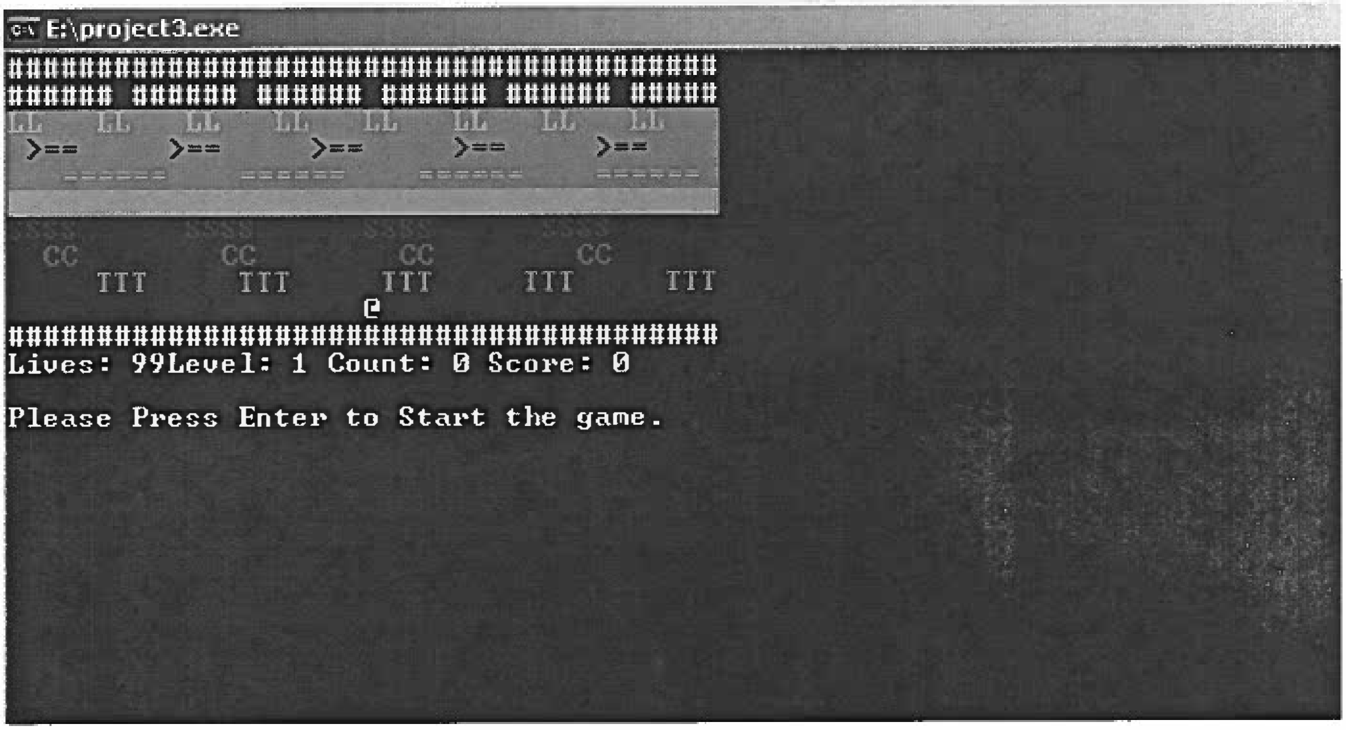
Kevin Rutkowski
COIN 496 Senior Portfolio
Monday, April 14, 2008

Designed for COIN 315
Grade Received- 100

The code was compiled under Visual C++ and Windows Vista. It uses keywords specific to Visual C++ (the include guards).


E:\project3.exe
Lives: 99Level: 1 Count: 0 Score: 0
Please Press Enter to Start the game.

```
//Student Name: Kevin Rutkowski
//  Program Name: Frogger
//  Creation Date:March 2008
//  Last Modified Date: March 2008
//  COIN Course: COINS 315
//  Grade Received: 100
//  Comments regarding design:
//  main focus of this project was inheritance-
//gator, falling log, shortlog inherit from water
// water inherits from road
//semitruck, car, tractor inherit from road
//road inherits from danger

// world runs the program telling everyone when to move
// after world has told a danger to move, it updates a collision table and determines if a danger
//has collided with a player or vice versa

#include <cstdlib>
#include "World.h"


void main(int argc, char *argv[] ) {
  int delay=50;
  if (argc==2) delay=atoi(argv[1]);
  World w(delay);
  w.run();
  return;
}
```

```
#pragma once
#include <string>
#include <conio.h>
#include <windows.h>

enum Color {black=0, blue=FOREGROUND_BLUE, green=FOREGROUND_GREEN, red=FOREGROUND_RED,
purple=FOREGROUND_BLUE + FOREGROUND_RED, cyan=FOREGROUND_BLUE+FOREGROUND_GREEN,
brown=FOREGROUND_GREEN+FOREGROUND_RED, white=FOREGROUND_BLUE+FOREGROUND_GREEN+FOREGROUND_RED};

void waitForEnter();// wait until the user presses ENTER
bool getInput(char& ch);

class CConsole {
public:
    CConsole(int width, int height);
    void setColor(Color fore, Color back) {SetConsoleTextAttribute(m_hConsole, fore + (back<<4) );}
    void waitForEnter();// wait until the user presses ENTER
    void gotoXY(int x, int y); //moves cursor to position x (from left), y (from top)
      //0,0 is upper left corner
    void printChar(int x, int y, char ch); //prints ch at x,y, moves cursor over 1 to right
    void printChar(char ch) {_putch(ch);} //prints ch where current cursor is, moves cursor over 1 to right
    void printString(int x, int y, const std::string& s);
    void printString(const std::string& s);
    void printStringClearLine(const std::string& s); //clears rest of line after string printed
    void printStringClearLine(int x, int y, const std::string& s);
    void printInt(int x, int y, int val); //print an integer value at x,y
    void printInt(int val); //print an integer value
    void delay(int amount) {Sleep(amount);}
private:
    HANDLE  m_hConsole;
    int     m_width;
    int     m_height;
};
```

```cpp
#include "CConsole.h"

void waitForEnter() {// wait until the user presses ENTER
  int ch;
  do {
    ch = _getch();
  } while ( ch != '\n'  &&  ch != '\r');
}

/* The getInput function checks whether the user has hit a key and immediately
 returns (so it does NOT wait for a key to be pressed).  It returns:
    true if the user has hit a key; in this case, the ch variable is set to
          the typed character
    false if the user has not hit any key; in this case, the ch variable
          will be unchanged*/
bool getInput(char& ch) {
  if (_kbhit()) {//has key been hit?
    int c = _getch();

    if (c != 0xE0)  // first of the two sent by arrow keys, E0 is escape
        ch = c;
    else {
      c = _getch(); //get second char sent by escape sequence from arrows
      switch (c) {
        case 'K':  /* left  */ ch = '4'; break;
        case 'M':  /* right */ ch = '6'; break;
        case 'H':  /* up    */ ch = '8'; break;
        case 'P':  /* down  */ ch = '2'; break;
        default:               ch = '?'; break;//special character but not arrow
      }
    }
    return true;
  }
  return false;
}

CConsole::CConsole(int width, int height) : m_width(width), m_height(height) {
  m_hConsole = GetStdHandle(STD_OUTPUT_HANDLE);
  CONSOLE_CURSOR_INFO x;
  GetConsoleCursorInfo(m_hConsole, &x);
  x.bVisible=false;
  SetConsoleCursorInfo(m_hConsole, &x);//hides the cursor
}

void CConsole::gotoXY(int x, int y) {//moves cursor to position x (from left), y (from top)
//0,0 is upper left corner
  if (x < 0  ||  x >= m_width  ||  y < 0  ||  y >= m_height)    //invalid x,y
      return;
  COORD pos = { x, y };//same as pos.X=x; pos.Y=y;
  SetConsoleCursorPosition(m_hConsole, pos);
}

void CConsole::printChar(int x, int y, char ch) {//prints ch at x,y, moves cursor over 1 to right
  gotoXY(x,y);
  printChar(ch);
}

void CConsole::printString(int x, int y, const std::string& s) {
  gotoXY(x,y);
  printString(s);
}

void CConsole::printString(const std::string& s) {
  for (size_t i = 0; i < s.length(); i++)
```

```
      printChar(s[i]);
}

void CConsole::printStringClearLine(int x, int y, const std::string& s) {//clears rest of line after string printed
  gotoXY(x,y);
  printStringClearLine(s);
}

void CConsole::printStringClearLine(const std::string& s) {//clears rest of line after string printed
  printString(s);
  CONSOLE_SCREEN_BUFFER_INFO info;
  GetConsoleScreenBufferInfo(m_hConsole,&info);
  for (int col=info.dwCursorPosition.X; col < m_width; col++)
      printChar(' ');//print spaces for rest of line, console b/c don't know if we started at 0
}

void CConsole::printInt(int x, int y, int val) {//print an integer value
  gotoXY(x,y);
  printInt(val);
}
void CConsole::printInt(int val) {//print an integer value
  char buf[12]={0};
  _itoa_s(val, buf, 11, 10);//safely convert int to ascii text
  printString(buf);  //constructor for string automatically called for char *
}
```

```
#pragma once

class CConsole;
#include <string>
using namespace std;
class Player;
class Danger;

const int COLUMN_SIZE=40;
const int ROW_SIZE=15;
const bool MOVES_RIGHT=true;
const int AMOUNT_OF_DANGERS=7;

class World{
public:

World(int);
~World();
void initialSetup();//set screen and collision table up to initial values
void movePlayer(char); //moves player and updates collision table
void run();// runs program
void drawBeach();// just draws the beach
int getLevel();// returns level
int increaseFrogsAtTop();//just increases frogs at top
int setLives();// reduce lives by 1
int getLives();// returns lives
int increaseLevel();//increases the level and displays message
void moveDangers();// moves dangers and updates collisiontable
int decreaseDelay();// reduce delay for leveling up
void updateCollisionTable(int,int,char);// updates collision table for the player taking xpos,ypos, and a char to place in the table as parameters
void updateCollisionTable(int);// updates collisiontable for dangers taking the dangers array index as a parameter
void setCollisionTableInitial();// sets the collision table to initial positions
void playerDied();// reset the game for when the player dies
void clearPreviousPositionInTable(int,int);
void resetWinningRow();//clears frogger from winning row
void movePlayerWithWater();// moves the player according to the waters direction
void titleScreen();
void askForReset();


private:
    CConsole * console;// pointer to console to access its methods
    Player * player;// pointer to player to access its methods
    Danger * arrOfDangers[AMOUNT_OF_DANGERS];   // array of danger pointers
    char collisionTable[ROW_SIZE][COLUMN_SIZE];// collisiontable used to determine if player dies
    int level;// level player is currently on
    int frogsAtTop;//how many frogs made it to the top
    int Delay;// delay of the game each tick
    int lives;// players lives
    int tick;// counts the iterations of the loop
    string winningRow;// string representation of row 1
};
```

```
#include "CConsole.h"
#include "World.h"
#include "Player.h"
#include "Danger.h"
#include "Road.h"
#include "SemiTruck.h"
#include "Car.h"
#include "Tractor.h"
#include "FallingLog.h"
#include "ShortLog.h"
#include "Gator.h"
#include "Snake.h"

World::World(int delay)
{
    system("cls");
    Delay = delay;
    console = new CConsole(COLUMN_SIZE, ROW_SIZE);
    player = new Player(console);
    arrOfDangers[0]= new Tractor(6,8,green,black,console, "      TTT      TTT      TTT      TTT      TTT",MOVES_RIGHT);//6
    arrOfDangers[1]= new Car(2,7,red,black,console, "  CC          CC          CC          CC       ",!MOVES_RIGHT);//2
    arrOfDangers[2]= new SemiTruck(3,6, blue, black, console, "SSSS       SSSS       SSSS       SSSS       ", MOVES_RIGHT);//3
    arrOfDangers[3]= new FallingLog(3,4,brown,cyan,console, "   ======     ======     ======     ====== ",MOVES_RIGHT);//3
    arrOfDangers[4]= new Gator(4,3,blue,cyan,console, " >==       >==       >==       >==       >==     ",!MOVES_RIGHT);//4
    arrOfDangers[5]= new ShortLog(3,2,brown,cyan,console, "LL    LL    LL    LL    LL    LL    LL    LL   ",MOVES_RIGHT);//3
    arrOfDangers[6]= new Snake(12,5,black,brown,console, "                                        ~~",!MOVES_RIGHT);
    level=1;
    frogsAtTop=0;
    lives=99;
    tick=1;
    for(int i=0; i<COLUMN_SIZE; i++)
    {
        if( (i+1) % 7 == 0)
            winningRow+=' ';
        else
            winningRow+='#';
    }
}

World::~World()
{
    delete console;
    delete player;
    for (int i=0; i<AMOUNT_OF_DANGERS; i++)
    {
        delete arrOfDangers[i];
    }
}
void World::initialSetup()
{
    for(int i=0; i < COLUMN_SIZE; i++)
    {
        console->printChar(i, 0, '#');
        console->printChar(i, 10, '#');
    }

    for(int i=0; i < AMOUNT_OF_DANGERS; i ++)
    {
        arrOfDangers[i]->initDraw();
```

```
    }
    console->printString(0, 1,winningRow);

    console->printString(0, 11, "Lives: ");
    console->printInt(7, 11, lives);

    console->printString(9, 11, "Level: ");
    console->printInt(16, 11, level);

    console->printString(18, 11, "Count: ");
    console->printInt(25, 11, frogsAtTop);

    console->printString(27, 11, "Score: ");

    player->displayScore();
    player->display();
    drawBeach();

    setCollisionTableInitial();
    updateCollisionTable(player->initialYposition(),player->initialXposition(),'@');

};


void World::run()// runs program
{
    char input;
    int dangerCount=AMOUNT_OF_DANGERS-1;
    titleScreen();
    system("cls");
    initialSetup();
    console->printString(0, 13,"Please Press Enter to Start the game.");
    waitForEnter();
    console->printStringClearLine(0, 13, " ");

    while(lives != 0)
    {
        console->delay(Delay);
        if(getInput(input))
        {
            movePlayer(input);
            player->displayScore();
            if(player->reachedTop())
            {
                increaseFrogsAtTop();
            }
        }
        input='Ì';
        moveDangers();

        tick++;
        if(tick==13)
        {
            tick=1;
        }
        for(int i=0; i <dangerCount; i++)
        {
            arrOfDangers[i]->drawString();
        }
        player->display();
        if(getLevel() >1)
            dangerCount=AMOUNT_OF_DANGERS;
        if(lives ==0)
            askForReset();
    }
};
```

```


void World::moveDangers()
{
    int dangerCount=AMOUNT_OF_DANGERS-1;
    if(getLevel() > 1)
            dangerCount=AMOUNT_OF_DANGERS;
    for(int i=0; i< dangerCount; i++)
    {
        if(tick%arrOfDangers[i]->getSpeed()== 0)
        {
            arrOfDangers[i]->draw();
            updateCollisionTable(i);
        }
    }
}
void World::updateCollisionTable(int i) // for dangers
{
    string dangerStr=arrOfDangers[i]->getNewString();
    int startCoord=arrOfDangers[i]->startCoordinate();
    bool movesRight=arrOfDangers[i]->getDirection();

    char templast=dangerStr[COLUMN_SIZE-1];// temp variables for first and last chars in the string
    char tempfirst=dangerStr[0];
    bool playerIsInRow=false;
    int playerPos=0;


    switch(startCoord)
    {
    // cases for water dangers
    case 3:
        for(int column=COLUMN_SIZE-1; column>-1;column--)
        {
            if(collisionTable[startCoord][column]=='@')
                playerIsInRow=true;

            collisionTable[startCoord][column]=dangerStr[column];
        }
        if(collisionTable[startCoord][0] =='@')
            playerIsInRow=true;

        collisionTable[startCoord][COLUMN_SIZE-1]=tempfirst;
        if(playerIsInRow)
            movePlayerWithWater();
        break;
    case 2:
    case 4:
        for(int column=1; column<COLUMN_SIZE;column++)
        {
            if(collisionTable[startCoord][column]=='@')
                playerIsInRow=true;

            collisionTable[startCoord][column]=dangerStr[column];
        }
        if(collisionTable[startCoord][0] =='@')
            playerIsInRow=true;

        collisionTable[startCoord][0]=templast;

        if(playerIsInRow)
            movePlayerWithWater();
        break;

    ///cases for road dangers
```

```
    case 7:
        for(int column=0; column<COLUMN_SIZE-1;column++)
        {
            if(collisionTable[startCoord][column] =='@')
            {
                playerIsInRow=true;
                playerPos=column;
            }
            collisionTable[startCoord][column]=dangerStr[column];
        }
        if(collisionTable[startCoord][0] =='@' && templast != ' ')
        {
            playerDied();
            return;
        }
        collisionTable[startCoord][COLUMN_SIZE-1]=tempfirst;
        if(playerIsInRow)
        {
            if(collisionTable[startCoord][playerPos] != ' ')
            {
                playerDied();
                return;
            }
            collisionTable[startCoord][playerPos]='@';
        }
        break;
    case 5:
    case 6:
    case 8:
        for(int column=1; column<COLUMN_SIZE;column++)
        {
            if(collisionTable[startCoord][column] =='@')
            {
                playerIsInRow=true;
                playerPos=column;
            }
            collisionTable[startCoord][column]=dangerStr[column];
        }
        if(collisionTable[startCoord][0] =='@')
        {
            playerIsInRow=true;
            playerPos=0;
        }
        collisionTable[startCoord][0]=templast;
        if(playerIsInRow)
        {
            if(collisionTable[startCoord][playerPos] != ' ')
            {
                playerDied();
                return;
            }
            collisionTable[startCoord][playerPos]='@';
        }
        break;
    default:break;
    }
}

int World::decreaseDelay(){return Delay-=5;}
int World::setLives(){return lives-=1;}
int World::getLives(){return lives;}
int World::getLevel(){return level;}
int World::increaseLevel()
{
    console->printInt(16, 11, level+=1);
    console->printInt(25, 11, frogsAtTop);
```

```

    console->printString(0, 12, "You have reached the next level!");
    console->printString(0, 13, "Please press enter to continue.");

    waitForEnter();

    console->printStringClearLine(0, 12," ");
    console->printStringClearLine(0, 13, " ");
    return level;
}
int World::increaseFrogsAtTop()
{
    frogsAtTop+=1;
    console->printInt(25, 11, frogsAtTop);
    player->setScore(level);
    player->reset();
    if(frogsAtTop == 5)
    {
        frogsAtTop=0;
        increaseLevel();
        decreaseDelay();
    }
    resetWinningRow();
    console->printString(0, 13, "You've reached the top!");
    console->printString(0,14,"Please press enter to continue.");
    waitForEnter();
    console->printStringClearLine(0,13," ");
    console->printStringClearLine(0,14," ");
    return frogsAtTop;
}
void World::drawBeach()
{
    for(int i =0; i < COLUMN_SIZE; i++)
    {
        console->setColor(white, brown);
        console->printChar(i,5, ' ');
    }
    console->setColor(white, black);
}
void World::setCollisionTableInitial()
{

    for (int i=0; i<AMOUNT_OF_DANGERS;i++)
    {
        string name=arrOfDangers[i]->initialStringPosition();
        for(int columns=0; columns<COLUMN_SIZE; columns++)
        {
            collisionTable[arrOfDangers[i]->startCoordinate()][columns]=name[columns]
    ;
            collisionTable[1][columns]=winningRow[columns];
        }
    }
    for(int i=0; i<COLUMN_SIZE;i++)
    {
        collisionTable[0][i]=' ';
        collisionTable[5][i]=' ';
        collisionTable[9][i]=' ';
    }
};
void World::movePlayer(char input)
{
    clearPreviousPositionInTable(player->getyLocation(), player->getxLocation());
    player->move(input);
    updateCollisionTable(player->getyLocation(), player->getxLocation(), '@');
};
void World::updateCollisionTable(int row,int col,char ch)
```

```
{
    switch (row)
    {
    case 1:
        if(collisionTable[row][col] != ' ')
        {
            playerDied();
            return;
        }
        break;
    case 2:
        if (col==0){}// dont check for player dieing at the end of the row
        else if(collisionTable[row][col] != 'L')
        {
            playerDied();
            return;
        }
        break;
    case 3:
        if (col==39){}//dont check for player dieing at end of the row
        else if(collisionTable[row][col] != '=')
            {
                playerDied();
                return;
            }
        break;
    case 4:
        if(collisionTable[row][col] == ' ' || col==39)
            {
                playerDied();
                return;
            }
        break;
    case 5:
    case 6:
    case 7:
    case 8:
            if(collisionTable[row][col] != ' ' && collisionTable[row][col] !='@')
            {
                playerDied();
                return;
            }
        break;
    default:break;
    }

    collisionTable[row][col]=ch;
};

void World::clearPreviousPositionInTable(int row,int col)
{
    switch(row)
    {
    case 2:collisionTable[row][col]='L';
        break;
    case 3:
    case 4:
        collisionTable[row][col]='=';
        break;
    default:collisionTable[row][col]=' ';
        break;
    }
};
void World::playerDied()
{
    setLives();
```

```
    player->reset();
    console->printString(0, 13,"You died! Press enter to continue.");
    waitForEnter();
    initialSetup();
    console->printStringClearLine(0,13," ");
    for(int i=0; i <AMOUNT_OF_DANGERS; i++)
    {
        arrOfDangers[i]->resetString();
    }
}
void World::resetWinningRow()
{
    console->printString(0, 1,winningRow);
    setCollisionTableInitial();
}

void World::movePlayerWithWater()
{
    char moveLeft='4';
    char moveRight='6';
    switch(player->getyLocation())
    {
    case 2:
    case 4: movePlayer(moveRight);
        break;
    case 3:movePlayer(moveLeft);
        break;
    default:break;
    }
};

void World::titleScreen()
{
    console->setColor(cyan,black);
    console->printString(0,3,"~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~↙
    ");
    console->setColor(green,black);
    console->printString(0,4,"||  //  ||==||  ||===||  ||===    ||===   ||===   ||==||  ↙
    ");
    console->printString(0,5,"|| //   ||  ||  ||   ||  ||       ||      ||      ||  ||  ↙
    ");
    console->printString(0,6,"||//    ||==||  ||   ||  ||       ||      ||===   ||==||  ↙
     ");
    console->printString(0,7,"||\\\\     ||\\\\     ||    ||  ||==||  ||==|| ||       ||\\↙
    \\  ");
    console->printString(0,8,"|| \\\\    || \\\\    ||    ||  ||  ||  ||   || ||       || \↙
    \\\ ");
    console->printString(0,9,"||  \\\\   ||  \\\\   ||===||  ||===    ||===   ||===   ||  ↙
    \\\\ ");
    console->setColor(cyan, black);
    console->printString(0,10,"~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~↙
    ~");


    char input=' ';
    int i=1;
    console->setColor(green,black);
    while(input!='\r')
    {

        console->delay(500);
        console->printString(0,12,"                              Press Enter to Continue"↙
    );
        console->delay(500);
        console->printString(0,12,"                                                      ↙
    ");
```

```
        i++;
        getInput(input);
    }
        console->setColor(white,black);
};

void World::askForReset()
{
    console->setColor(purple,black);
    console->printString(0,13,"Press enter to play again");
    console->printString(0,14,"or push any other key to quit.");
    console->setColor(white,black);
    char input=' ';
    while(input == ' ')
    {
        getInput(input);
        if(input == '\r')
        {
            lives=5;
            initialSetup();
        }
    }
    input=' ';

    console->printStringClearLine(0,13," ");
    console->printStringClearLine(0,14," ");
}
```

```
#include "Player.h"
class CConsole;

Player::Player(CConsole* C)
{
    console = C;
    xLoc= 20;
    yLoc= 9;
    score= 0;
    maxHeight=yLoc;
    atTop=false;
};

void Player::clearPrevious()
{
    int y=getyLocation();

    switch(y)
    {
        case 2:
            console->setColor(brown, cyan);
            console->printChar(getxLocation(), getyLocation(), ' ');
            break;
        case 3:
            console->setColor(blue, cyan);
            console->printChar(getxLocation(), getyLocation(), '=');
            break;
        case 4:
            console->setColor(brown, cyan);
            console->printChar(getxLocation(), getyLocation(), '=');
            break;
        default:
            console->printChar(getxLocation(), getyLocation(), ' ');
            break;
        case 5:
            console->setColor(white, brown);
            console->printChar(getxLocation(), getyLocation(), ' ');
            break;
    }
    console->setColor(white, black);
};
void Player::display()
{
    switch(getyLocation())
    {
        case 2:
        case 3:
        case 4:
            console->setColor(white, cyan);
            console->printChar(getxLocation(), getyLocation(), '@');
            console->setColor(white, black);
                break;
        case 5:
            console->setColor(white, brown);
            console->printChar(getxLocation(), getyLocation(), '@');
            console->setColor(white, black);
                break;
        default:
            console->printChar(getxLocation(), getyLocation(), '@');
                break;
    }


};

void Player::reset()
```

```
{
    atTop=false;
    setX(getxLocation() * -1);
    setX(20);
    setY(getyLocation() * -1);
    setY(9);
    console->printChar(getxLocation(), getyLocation(), '@');
    resetMaxHeight();
};

int Player::setX(int val)
{
    switch(xLoc+=val)
    {
    case -1: return xLoc=39; break;
    case 40: return xLoc=0; break;
    default: return xLoc; break;
    }

};
int Player::setY(int val)
{
    switch(yLoc+=val)
    {
    case 0: return yLoc=1; break;
    case 1:
        {
            setScore();
            atTop=true;
            return 9;
            break;
        }
    case 10: return yLoc=9; break;
    default:
        {
            if(yLoc < maxHeight)
            {
                maxHeight = yLoc;
                setScore();
            }
            return yLoc; break;
        }
    }
};
int Player::getxLocation(){return xLoc;}
int Player::getyLocation(){return yLoc;}
int Player::setScore(){return score+=5;}
int Player::setScore(int level){return score=score+level*50;}
int Player::displayScore()
{
    console->printInt(34, 11, score);
    return score;
}

void Player::move(char input)
{
    switch(input)
    {
    case '4':
        clearPrevious();
        setX(-1);//left
        display();
        break;
    case '6':
        clearPrevious();
        setX(1);//right
```

```
        display();
        break;
    case '8':
        clearPrevious();
        setY(-1);//up
        display();
        break;
    case '2':
        clearPrevious();
        setY(1);//down
        display();
        break;
    }
};

bool Player::reachedTop(){return atTop;}
int Player::initialXposition(){return xLoc=20;}
int Player::initialYposition(){return yLoc=9;}
int Player::resetMaxHeight(){return maxHeight=9;}
```

```
#pragma once
#include "CConsole.h"
#include "World.h"


class Player{
public:
    Player(CConsole*);
    void clearPrevious();// clears previous position on screen
    void display(); //displays player
    void move(char);// moves the player
    void reset();//resets player  for when he dies/gets to the next level
    int getxLocation();// returns players xlocation on the screen
    int getyLocation();// returns players ylocation on the screen
    int setX(int);// set players x position by taking an int as a parameter to add to current position
    int setY(int);// set players x position by taking an int as a parameter to add to current position
    int initialXposition();// save x initial position
    int initialYposition();// save y initial position
    int setScore();// change score as players moves
    int setScore(int);// change score as player advances a level
    int displayScore();// displays score
    bool reachedTop();// determines if player has reached top
    int resetMaxHeight();// resests the max height variable to score player correctly


private:
    int xLoc;// players xloc
    int yLoc;//players y loc
    int maxHeight;//max height player has reached for the current life
    int score;// score
    bool atTop;//true if at top false otherwise
    CConsole *console;//pointer to access cconsoles methods
};
```

```
#pragma once
#include "CConsole.h"
#include "World.h"

class Danger{
public:
Danger(string,int);
int getSpeed();// returns the speed of the danger
string getName();// returns the name of the danger
virtual void initDraw()=0;// draws the dangeres in their initial position
virtual void draw()=0;// draws dangers as they move
virtual int startCoordinate()=0;//returns which row the danger starts on
virtual bool getDirection()=0;//returns true if the danger moves right false if left
virtual string initialStringPosition()=0;// the initial value of the string to reset the screen/collisiontable
virtual string getNewString()=0;//updated string as it moves
virtual string resetString()=0;//resets the getNewString to the initial positions
virtual void drawString()=0;


protected:
    string dangerName;
    int speed;
};
```

```
class CConsole;
#include "Danger.h"

Danger::Danger(string dangerStr,int vel)
{
    dangerName=dangerStr;
    speed=vel;
};

int Danger::getSpeed()
{
    return speed;
}
string Danger::getName()
{
    return dangerName;
}

```

```
#pragma once
#include "Danger.h"
class Road:public Danger
{
public:
    Road(int,int,Color,Color,CConsole*,string,bool);
    virtual void initDraw();// draws initial strings on screen
    void draw();// updates the screen with the strings new positions
    int startCoordinate();// returns row danger is on
    bool getDirection();// returns true if the danger moves right false otherwise
    string initialStringPosition();/// the initial value of the string to reset the
    screen/collisiontable
    string getNewString();//updated string as it moves
    string resetString();//resets the getNewString to the initial positions
    void drawString();// prints the strings to the screen

private:
    CConsole *console;
    string vehicle;
    string initialStringPos;
    string oldVehicle;
    Color fore;
    Color back;
    int startCoord;
    bool direction;
};
```

```
#include "Road.h"

Road::Road(int vel,int start, Color forcolor, Color backcolor, CConsole* con,string dangerName, bool dir):Danger(dangerName, vel)
{
    console =con;
    vehicle=dangerName;
    oldVehicle=dangerName;
    initialStringPos=dangerName;
    fore=forcolor;
    back=backcolor;
    startCoord=start;
    direction=dir;
}
void Road::initDraw()
{
    console->setColor(fore, back);
    console->printString(0,startCoord, dangerName);
    console->setColor(white, black);
};

void Road::draw()
{
    char tempfront=vehicle[0];
    char tempback=vehicle[39];
    oldVehicle=vehicle;
    if(direction==false)
    {
        vehicle.erase(0,1);
        vehicle.push_back(tempfront);
    }
    else
    {
        vehicle.erase(39,1);
        vehicle=tempback+vehicle;
    }
}

int Road::startCoordinate(){return startCoord;}
bool Road::getDirection(){return direction;}
string Road::initialStringPosition(){return initialStringPos;}
string Road::getNewString(){return vehicle;}
string Road::resetString(){return vehicle=initialStringPosition();}

void Road::drawString()
{
    console->setColor(fore, back);
    //console->printString(0, startCoord, vehicle);

    for(int i=0; i< (signed)vehicle.length();i++)
    {
        if(vehicle[i] == oldVehicle[i] && startCoordinate() != 2)
        {
            continue;
        }//don't print to prevent flashing
        else
        {
            console->printChar(i, startCoordinate(), vehicle[i]);
        }
    }
    console->setColor(white, black);
}
```

```
#pragma once
#include "Road.h"

class Car:public Road
{
public:
    Car(int,int,Color,Color,CConsole*,string dangerName, bool);
private:

};
```

```
#include "Car.h"

Car::Car(int vel,int startCoord, Color forcolor, Color backcolor, CConsole* con,string dangerName, bool dir):Road(vel,startCoord, forcolor,backcolor,con,dangerName, dir)
{
}
```

```
#pragma once
#include "Road.h"

class SemiTruck:public Road
{
public:
    SemiTruck(int,int,Color,Color,CConsole*,string dangerName, bool);
private:

};
```

```
#include "SemiTruck.h"

SemiTruck::SemiTruck(int vel,int startCoord, Color forcolor, Color backcolor, CConsole*
    con,string dangerName, bool dir):Road(vel,startCoord, forcolor,backcolor,con,
    dangerName, dir)
{
}
```

```
#pragma once
#include "Road.h"

class Snake:public Road
{
public:
    Snake(int,int,Color,Color,CConsole*,string, bool);
private:

};
```

```
#include "Snake.h"

Snake::Snake(int vel,int startCoord, Color forcolor, Color backcolor, CConsole* con, string dangerName, bool dir):Road(vel,startCoord, forcolor,backcolor,con,dangerName, dir)
{
}
```

```
#pragma once
#include "Road.h"

class Tractor:public Road
{
public:
    Tractor(int,int,Color,Color,CConsole*,string, bool);
private:

};
```

```
#include "Tractor.h"

Tractor::Tractor(int vel,int startCoord, Color forcolor, Color backcolor, CConsole* con
    ,string dangerName, bool dir):Road(vel,startCoord, forcolor,backcolor,con,
    dangerName, dir)
{
}
```

```
#pragma once
#include "Road.h"
class Water:public Road
{
public:
    Water(int,int,Color,Color,CConsole*,string,bool);

private:
};
```

```
#include "Water.h"

Water::Water(int vel,int startCoord, Color forcolor, Color backcolor, CConsole* con, string dangerName, bool dir):Road(vel,startCoord, forcolor,backcolor,con,dangerName, dir)
{}
```

```
#pragma once
#include "Water.h"

class FallingLog:public Water
{
public:
    FallingLog(int,int,Color,Color,CConsole*,string, bool);
private:

};
```

```
#include "FallingLog.h"

FallingLog::FallingLog(int vel,int startCoord, Color forcolor, Color backcolor, CConsole* con,string dangerName, bool dir):Water(vel,startCoord, forcolor,backcolor,con,dangerName, dir)
{}
```

```
#pragma once
#include "Water.h"

class Gator:public Water
{
public:
    Gator(int,int,Color,Color,CConsole*,string, bool);
private:

};
```

```
#include "Gator.h"

Gator::Gator(int vel,int startCoord, Color forcolor, Color backcolor, CConsole* con, string dangerName, bool dir):Water(vel,startCoord, forcolor,backcolor,con, dangerName, dir)
{}
```

```
#pragma once
#include "Water.h"

class ShortLog:public Water
{
public:
    ShortLog(int,int,Color,Color,CConsole*,string, bool);
private:

};
```

```
#include "ShortLog.h"

ShortLog::ShortLog(int vel,int startCoord, Color forcolor, Color backcolor, CConsole* con,string dangerName, bool dir):Water(vel,startCoord, forcolor,backcolor,con, dangerName, dir)
{}
```

Designed for COIN 325
Grade Received- 100

The code was compiled under Visual J# and Windows Vista. Enter a phrase or word and the program pushes the word onto two stacks, one from front to back the other from back to front. It then pops each character off and compares them to tell if the word is a palindrome.

Command Prompt

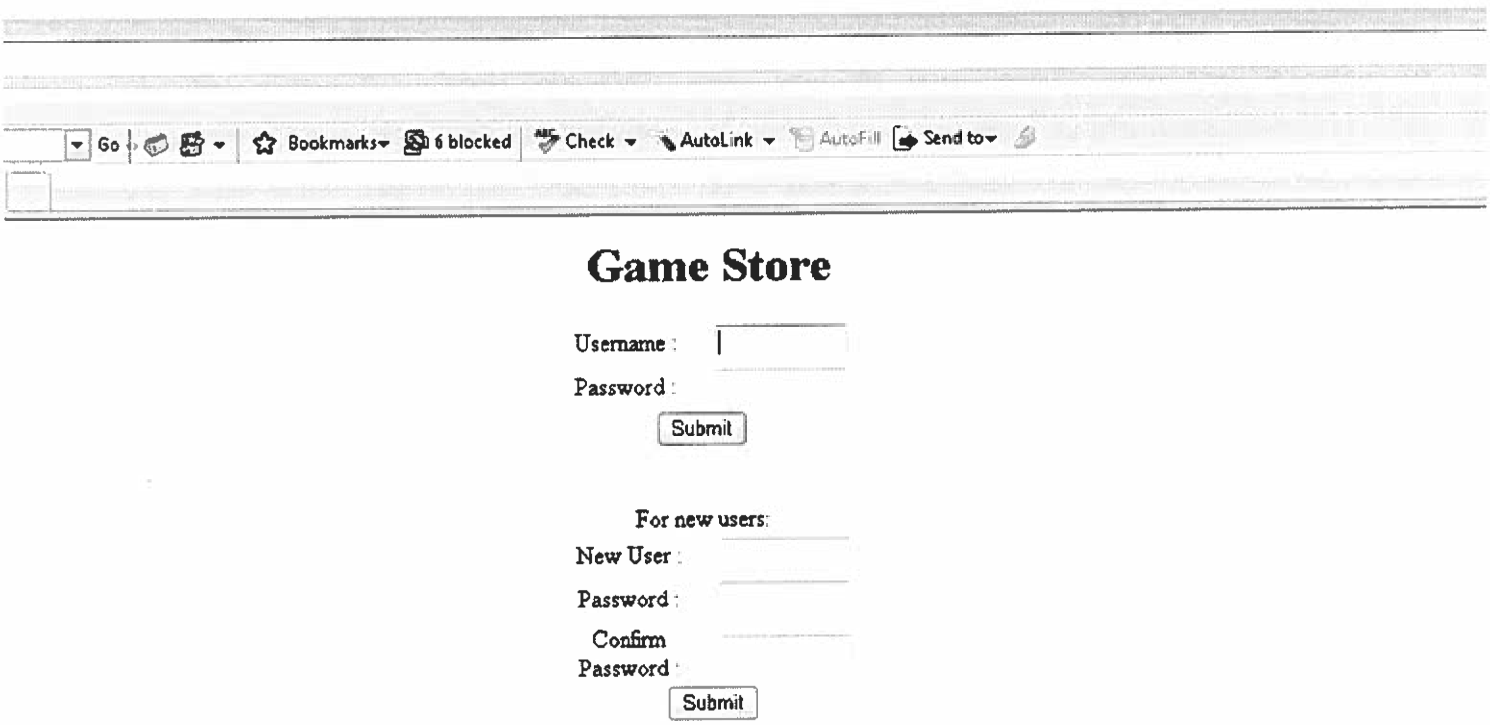
```
Microsoft Windows [Version 5.2.3790]
(C) Copyright 1985-2003 Microsoft Corp.

x:\>java stack
Please enter a word or phrase and the program will decide if its a palindrome:
madam i'm adam
[m, a, d, a, m, i, m, a, d, a, m]
[m, a, d, a, m, i, m, a, d, a, m]
This is a palindrome.

x:\>
```

```
//Student Name: Kevin Rutkowski
//  Program Name: Palindrome
//  Creation Date: Fall 2007
//  Last Modified Date: Fall 2007
//  COIN Course: COINS 325
//  Grade Received: 100
//  Comments regarding design:
// goal of the program is to determine if a string is a palindrome by using stacks
    ignoring spaces
// user inputs string
// program then puts the string on a stack

import java.util.Stack;
import java.util.Scanner;

public class stack
{
    public static void main(String args[])
    {
        Stack stack = new Stack();
        Stack stack1 = new Stack();
        Scanner input = new Scanner(System.in);
        boolean palin = true;


        System.out.println("Please enter a word or phrase and the program will decide if
    its a palindrome: ");
        String word = input.nextLine();
        int length = word.length();

        // push the characters on the string onto the stack if they aren't a puncuation
    or space
        for (int i = 0; i < word.length(); i++)
        {
            length -= 1;

            if (Character.isLetter(word.charAt(i)))
            {
                stack.push(word.charAt(i));
            }
            if (Character.isLetter(word.charAt(length)))
            {
                stack1.push(word.charAt(length));
            }

        }
        //output the contents of both stacks to verify
        System.out.println(stack);
        System.out.println(stack1);

        while (!stack.empty())
        {
            if (stack1.peek() == stack.peek())
            {
                stack.pop();
                stack1.pop();
            }
            else
            {
                System.out.println("This is not a palindrome.");
                palin = false;
                break;
            }
        }
        if (palin == true)
        {
```

```
            System.out.println("This is a palindrome.");
        }

    }
}
```

Designed for COIN 332
Grade received-100

This was our term project. We had to Design a website that could sell products. User could create an account or log into their own account letting the webpage do all the database checking.


Go
Bookmarks
6 blocked
Check
AutoLink
AutoFill
Send to
Game Store
Username :
Password :
Submit
For new users:
New User :
Password :
Confirm
Password :
Submit

```
<%@ Page Language="VB" AutoEventWireup="false" CodeFile="store.aspx.vb" Inherits="store" ↙
    %>

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/     ↙
    xhtml1/DTD/xhtml1-transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml" >
<head runat="server">
    <title>Untitled Page</title>
</head>
<body>
    <form id="form1" runat="server">
    <div>
    <center><h1>
        Game Store</h1>
        <p>
            <table>
                <tr>
                    <td style="width: 100px">
                    <img src="cod.jpg" alt=""/>
                    </td>
                    <td style="width: 100px">
                        Call of Duty 4</td>
                    <td style="width: 100px">
                        $59.99<asp:CheckBox ID="cod" runat="server" /></td>
                </tr>
                <tr>
                    <td style="width: 100px">
                        <img src="GTA.jpg" /></td>
                    <td style="width: 100px">
                        Grand Theft Auto 4</td>
                    <td style="width: 100px">
                        $59.99<asp:CheckBox ID="gta" runat="server" /></td>
                </tr>
                <tr>
                    <td style="width: 100px">
                        <img src="halo.jpg" /></td>
                    <td style="width: 100px">
                        Halo 3</td>
                    <td style="width: 100px">
                        $59.99<asp:CheckBox ID="h3" runat="server" /></td>
                </tr>
                <tr>
                    <td style="width: 100px">
                        <img src="halose.jpg" /></td>
                    <td style="width: 100px">
                        Halo 3: Special Edition</td>
                    <td style="width: 100px">
                        $69.99<asp:CheckBox ID="h3se" runat="server" /></td>
                </tr>
                <tr>
                    <td style="width: 100px">
                        <img src="halol.jpg" /></td>
                    <td style="width: 100px">
                        Halo 3: Legendary Edition</td>
                    <td style="width: 100px">
                        $89.99<asp:CheckBox ID="h3le" runat="server" /></td>
                </tr>
                <tr>
                    <td style="width: 100px">
                        <img src="headset.jpg" /></td>
                    <td style="width: 100px">
                        Headset</td>
                    <td style="width: 100px">
                        $59.99<asp:CheckBox ID="headset" runat="server" /></td>
                </tr>
```

```
            <tr>
                <td style="width: 100px">
                    <img src="controller.jpg" /></td>
                <td style="width: 100px">
                    Controller</td>
                <td style="width: 100px">
                    $32.99<asp:CheckBox ID="controller" runat="server" /></td>
            </tr>
            <tr>
                <td style="width: 100px">
                    <img src="xbox.jpg" /></td>
                <td style="width: 100px">
                    Xbox 360</td>
                <td style="width: 100px">
                    $399.99<asp:CheckBox ID="xbox" runat="server" /></td>
            </tr>
            <tr>
                <td style="width: 100px">
                    <img src="xboxlive.jpg" /></td>
                <td style="width: 100px">
                    Xbox Live Membership</td>
                <td style="width: 100px">
                    $50.00<asp:CheckBox ID="live" runat="server" /></td>
            </tr>
            <tr>
                <td style="width: 100px">
                    <img src="gow.jpg" /></td>
                <td style="width: 100px">
                    Gears Of War</td>
                <td style="width: 100px">
                    $59.99<asp:CheckBox ID="gow" runat="server" /></td>
            </tr>
        </table>
    </p>
    <p>
        <asp:Button ID="purchase" runat="server" Text="Purchase" /> <asp:
SqlDataSource
            ID="SqlDataSource1" runat="server" ConnectionString="<%$
ConnectionStrings:ConnectionString %>"
            DeleteCommand="DELETE FROM [database] WHERE [id] = @id" InsertCommand=
"INSERT INTO [database] ([name], [password], [cod4], [gta4], [halo3], [halo3se],
[halo3le], [headset], [controller], [xboxlive], [xbox], [gow]) VALUES (@name, @
password, @cod4, @gta4, @halo3, @halo3se, @halo3le, @headset, @controller, @xboxlive,
 @xbox, @gow)"
            SelectCommand="SELECT * FROM [database]" UpdateCommand="UPDATE [database]
 SET [name] = @name, [password] = @password, [cod4] = @cod4, [gta4] = @gta4, [halo3]
= @halo3, [halo3se] = @halo3se, [halo3le] = @halo3le, [headset] = @headset,
[controller] = @controller, [xboxlive] = @xboxlive, [xbox] = @xbox, [gow] = @gow
WHERE [id] = @id">
            <DeleteParameters>
                <asp:Parameter Name="id" Type="Int32" />
            </DeleteParameters>
            <UpdateParameters>
                <asp:Parameter Name="name" Type="String" />
                <asp:Parameter Name="password" Type="String" />
                <asp:Parameter Name="cod4" Type="String" />
                <asp:Parameter Name="gta4" Type="String" />
                <asp:Parameter Name="halo3" Type="String" />
                <asp:Parameter Name="halo3se" Type="String" />
                <asp:Parameter Name="halo3le" Type="String" />
                <asp:Parameter Name="headset" Type="String" />
                <asp:Parameter Name="controller" Type="String" />
                <asp:Parameter Name="xboxlive" Type="String" />
                <asp:Parameter Name="xbox" Type="String" />
                <asp:Parameter Name="gow" Type="String" />
                <asp:Parameter Name="id" Type="Int32" />
```

```
                </UpdateParameters>
                <InsertParameters>
                    <asp:Parameter Name="name" Type="String" />
                    <asp:Parameter Name="password" Type="String" />
                    <asp:Parameter Name="cod4" Type="String" />
                    <asp:Parameter Name="gta4" Type="String" />
                    <asp:Parameter Name="halo3" Type="String" />
                    <asp:Parameter Name="halo3se" Type="String" />
                    <asp:Parameter Name="halo3le" Type="String" />
                    <asp:Parameter Name="headset" Type="String" />
                    <asp:Parameter Name="controller" Type="String" />
                    <asp:Parameter Name="xboxlive" Type="String" />
                    <asp:Parameter Name="xbox" Type="String" />
                    <asp:Parameter Name="gow" Type="String" />
                </InsertParameters>
            </asp:SqlDataSource>
        </p>
    </center>
     </div>
    </form>
</body>
</html>
```

```
Imports System.Data.Sql
Imports System.Data
Partial Class _Default
    Inherits System.Web.UI.Page

    Dim verified As Boolean

    Protected Sub submit_Click(ByVal sender As Object, ByVal e As System.EventArgs)  ↙
    Handles submit.Click
        incorrect.Text = ""
        Dim user As String
        Dim newpass As String
        Dim confirmpass As String
        'set up opening commands
        Dim conn As New System.Data.SqlClient.SqlConnection(ConfigurationManager.  ↙
    ConnectionStrings("DatabaseConnectionString").ToString())
        Dim cmd As New System.Data.SqlClient.SqlCommand
        Dim check As New System.Data.SqlClient.SqlCommand
        conn.Open()
        user = newuser.Text
        newpass = newpassword.Text
        confirmpass = confirmpassword.Text


        check = New System.Data.SqlClient.SqlCommand("SELECT COUNT(*)FROM [database]WHERE↙
     (name = '" + user + "')", conn)

        Dim result As Integer = check.ExecuteScalar

        If user <> "" And newpass = "" And confirmpass <> "" Then
            If result = 1 Then
                incorrect.Text = "Username is already in use."
            Else
                If newpass = confirmpass Then
                    Dim Sql As String
                    Sql = "INSERT INTO [database] ([name], [password]) VALUES ('" + user ↙
    + "','" + newpass + "')"
                    cmd = New SqlClient.SqlCommand(Sql, conn)
                    cmd.ExecuteNonQuery()
                    Session("name") = user
                    Response.Redirect("store.aspx?name='" & newuser.Text & "'")
                Else
                    incorrect.Text = "Passwords do not match."
                End If
            End If
            conn.Close()
        Else
            incorrect.Text = "Please enter a value in all fields."
        End If

    End Sub


    Protected Sub Page_Load(ByVal sender As Object, ByVal e As System.EventArgs) Handles ↙
    Me.Load


    End Sub

    Protected Sub SqlDataSource1_Selecting(ByVal sender As Object, ByVal e As System.Web.↙
    UI.WebControls.SqlDataSourceSelectingEventArgs) Handles SqlDataSource1.Selecting

    End Sub

    Protected Sub Button1_Click(ByVal sender As Object, ByVal e As System.EventArgs)  ↙
    Handles Button1.Click
```

```
        require.Text = ""
        Dim conn As New System.Data.SqlClient.SqlConnection(ConfigurationManager. ↙
    ConnectionStrings("DatabaseConnectionString").ToString())
        Dim command As New System.Data.SqlClient.SqlCommand
        Dim user As String
        Dim pass As String
        conn.Open()
        user = username.Text
        pass = password.Text

        Dim check As New System.Data.SqlClient.SqlCommand
        check = New System.Data.SqlClient.SqlCommand("SELECT COUNT(*)FROM [database]WHERE↙
     (name = '" + user + "')", conn)
        Dim result As Integer = check.ExecuteScalar

        If result = 0 Then
            require.Text = "No such username."
        Else
            command = New System.Data.SqlClient.SqlCommand("SELECT * FROM [database]    ↙
    WHERE name ='" + user + "'", conn)
            Dim name As String = ""
            Dim password As String = ""
            Dim read As System.Data.SqlClient.SqlDataReader = command.ExecuteReader()

            While read.Read
                name = read.Item("name")
                password = read.Item("password")
            End While
            'remove any spaces that the datbase recovered from the end
            name = name.TrimEnd()
            password = password.TrimEnd()

            If pass <> password Then
                require.Text = "Incorrect password or Username."
            Else
                Session("name") = user
                Response.Redirect("store.aspx?name='" & user & "'")
            End If
            read.Close()
            conn.Close()
        End If
    End Sub

    Protected Sub newuser_TextChanged(ByVal sender As Object, ByVal e As System.      ↙
    EventArgs) Handles newuser.TextChanged

    End Sub
End Class
```

```
<%@ Page Language="VB" AutoEventWireup="false" CodeFile="home.aspx.vb" Inherits="_Default↙
    " %>
<%@ Import Namespace="System.Data.Sql" %>
<%@ Import Namespace="System.Data" %>

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/     ↙
    xhtml1/DTD/xhtml1-transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml" >
<head runat="server">
    <title>Home</title>
    <script type="text/javascript" language="javascript">
    <!--
    // for some reason i can't get my button click to enter this
        function newuser()
    {
        var name = document.getElementById("newuser")
        var pass = document.getElementById("newpassword")
        var pass2 = document.getElementById("confirmpassword")
        alert("in")// to test if i get inside the function
        if (name.value == "")
        {
            alert("Please enter a username")
            name.focus();
            name.select();
            return false;
        }
        if (pass.value == "" || pass2.value == "")
        {
            alert("Please enter a password in each text box.")
            pass.focus();
            pass.select();
            return false;
        }
        if(pass.value != pass2.value)
        {
            alert("Passwords do not match\n Please re-enter.")
            pass.focus();
            pass.select();
            return false;
        }
        return true;
    }
     function passcheck()
    {
         var name = document.getElementById("username")
         var pass = document.getElementById("password")
         if (name.value == "")
         {
            alert("Please Enter a username")
            name.focus();
            name.select();
            return false;
         }
         if(pass.value == "")
         {
            alert("Please enter a password.")
            pass.focus();
            pass.select();
            return false;
         }
        return true;
        }


    -->
```

```
    </script>
    </head>
<body >
    <form id="form1" runat="server">
    <div>
    <center><h1>
         Game Store</h1></center>
        <center>
            <table>
                <tr>
                    <td style="width: 100px">
        <asp:Label ID="Label1" runat="server" Text="Username :"></asp:Label></td>
                    <td style="width: 100px">
        <asp:TextBox ID="username" runat="server" Columns="9"></asp:TextBox></td>
                </tr>
                <tr>
                    <td style="width: 100px; height: 26px;">
            <asp:Label ID="Label2" runat="server" Text="Password :"></asp:Label></td>
                    <td style="width: 100px; height: 26px;">
                        <asp:TextBox ID="password" runat="server" Columns="9" TextMode= ↙
    "Password"></asp:TextBox></td>
                </tr>
            </table>
        </center>
        <center>
            <asp:Button ID="Button1" runat="server" Text="Submit" onClientClick="return ↙
    passcheck()"/> </center>
        <center>
            <asp:Label ID="require" runat="server" Font-Bold="True" ForeColor="Red"></asp↙
    :Label> </center>
        <center>
             </center>
        <center>
            <asp:Label ID="Label6" runat="server" Text="For new users:"></asp:Label> ↙
    ;</center>
        <center>
            <table>
                <tr>
                    <td style="width: 100px">
            <asp:Label ID="Label3" runat="server" Text="New User :"></asp:Label></td>
                    <td style="width: 99px">
                        <asp:TextBox ID="newuser" runat="server" Columns="9"></asp:       ↙
    TextBox></td>
                </tr>
                <tr>
                    <td style="width: 100px; height: 26px;">
            <asp:Label ID="Label4" runat="server" Text="Password :"></asp:Label></td>
                    <td style="width: 99px; height: 26px;">
                        <asp:TextBox ID="newpassword" runat="server" Columns="9" TextMode↙
    ="Password"></asp:TextBox></td>
                </tr>
                <tr>
                    <td style="width: 100px">
            <asp:Label ID="Label5" runat="server" Text="Confirm Password :"></asp:Label> ↙
    </td>
                    <td style="width: 99px">
                        <asp:TextBox ID="confirmpassword" runat="server" Columns="9"     ↙
    TextMode="Password"></asp:TextBox></td>
                </tr>
            </table>
             
            <asp:Button ID="submit" runat="server" Text="Submit"/>
            </center>
        <center>
            <asp:Label ID="incorrect" runat="server" Font-Bold="True" ForeColor="Red"></ ↙
    asp:Label> </center>
```

```
        <center>
            <asp:SqlDataSource ID="SqlDataSource1" runat="server" ConnectionString="<%$ 
    ConnectionStrings:DatabaseConnectionString %>"
                DeleteCommand="DELETE FROM [database] WHERE [id] = @id"
                InsertCommand="INSERT INTO [database] ([name], [password]) VALUES (@name, 
     @password)"
                SelectCommand="SELECT * FROM [database]" UpdateCommand="UPDATE [database] 
     SET [name] = @name, [password] = @password, [cod4] = @cod4, [gta4] = @gta4, [halo3] 
    = @halo3, [halo3se] = @halo3se, [halo3le] = @halo3le, [headset] = @headset, 
    [controller] = @controller, [xboxlive] = @xboxlive, [xbox] = @xbox, [gow] = @gow 
    WHERE [id] = @id" ProviderName="<%$ ConnectionStrings:DatabaseConnectionString.
    ProviderName %>">
                <DeleteParameters>
                    <asp:Parameter Name="id" Type="Int32" />
                </DeleteParameters>
                <UpdateParameters>
                    <asp:Parameter Name="name" Type="String" />
                    <asp:Parameter Name="password" Type="String" />
                    <asp:Parameter Name="cod4" Type="String" />
                    <asp:Parameter Name="gta4" Type="String" />
                    <asp:Parameter Name="halo3" Type="String" />
                    <asp:Parameter Name="halo3se" Type="String" />
                    <asp:Parameter Name="halo3le" Type="String" />
                    <asp:Parameter Name="headset" Type="String" />
                    <asp:Parameter Name="controller" Type="String" />
                    <asp:Parameter Name="xboxlive" Type="String" />
                    <asp:Parameter Name="xbox" Type="String" />
                    <asp:Parameter Name="gow" Type="String" />
                    <asp:Parameter Name="id" Type="Int32" />
                </UpdateParameters>
                <InsertParameters>
                    <asp:Parameter Name="name" Type="String" />
                    <asp:Parameter Name="password" Type="String" />
                </InsertParameters>
            </asp:SqlDataSource>
             </center>
        <center>
             </center>

    </div>
    </form>
</body>
</html>
```

```
Partial Class store
    Inherits System.Web.UI.Page

    Protected Sub Page_Load(ByVal sender As Object, ByVal e As System.EventArgs) Handles ↙
    Me.Load
        Dim name As String = Request.QueryString("name")
        Session("cod") = False
        Session("gta4") = False
        Session("h3") = False
        Session("h3se") = False
        Session("h3le") = False
        Session("headset") = False
        Session("controller") = False
        Session("xbox") = False
        Session("xboxlive") = False
        Session("gow") = False
    End Sub

    Protected Sub purchase_Click(ByVal sender As Object, ByVal e As System.EventArgs)     ↙
    Handles purchase.Click
        Dim conn As New System.Data.SqlClient.SqlConnection(ConfigurationManager.         ↙
    ConnectionStrings("DatabaseConnectionString").ToString())
        Dim command As New System.Data.SqlClient.SqlCommand
        Dim item As String
        Dim amount As Integer = 0
        Dim name As String = Session("name")
        conn.Open()
        Dim total As Double = 0
        If cod.Checked Then
            total += 59.99
            item = "cod4"
            command = New System.Data.SqlClient.SqlCommand("UPDATE [database] SET " +     ↙
    item + "=" + "'true'" + "WHERE (name ='" + name + "')", conn)
            command.ExecuteNonQuery()
            Session("cod") = True
            amount += 1
        End If
        If gta.Checked Then
            total += 59.99
            item = "gta4"
            command = New System.Data.SqlClient.SqlCommand("UPDATE [database] SET " +     ↙
    item + "=" + "'true'" + "WHERE (name ='" + name + "')", conn)
            command.ExecuteNonQuery()
            Session("gta") = True
            amount += 1
        End If
        If h3.Checked Then
            total += 59.99
            item = "halo3"
            command = New System.Data.SqlClient.SqlCommand("UPDATE [database] SET " +     ↙
    item + "=" + "'true'" + "WHERE (name ='" + name + "')", conn)
            command.ExecuteNonQuery()
            Session("h3") = True
            amount += 1
        End If
        If h3se.Checked Then
            total += 69.99
            item = "halo3se"
            command = New System.Data.SqlClient.SqlCommand("UPDATE [database] SET " +     ↙
    item + "=" + "'true'" + "WHERE (name ='" + name + "')", conn)
            command.ExecuteNonQuery()
            Session("h3se") = True
            amount += 1
        End If
        If h3le.Checked Then
```

```
            total += 89.99
            item = "halo3le"
            command = New System.Data.SqlClient.SqlCommand("UPDATE [database] SET " +
    item + "=" + "'true'" + "WHERE (name ='" + name + "')", conn)
            command.ExecuteNonQuery()
            Session("h3le") = True
            amount += 1
        End If
        If headset.Checked Then
            total += 59.99
            item = "headset"
            command = New System.Data.SqlClient.SqlCommand("UPDATE [database] SET " +
    item + "=" + "'true'" + "WHERE (name ='" + name + "')", conn)
            command.ExecuteNonQuery()
            Session("headset") = True
            amount += 1
        End If
        If controller.Checked Then
            total += 32.99
            item = "controller"
            command = New System.Data.SqlClient.SqlCommand("UPDATE [database] SET " +
    item + "=" + "'true'" + "WHERE (name ='" + name + "')", conn)
            command.ExecuteNonQuery()
            Session("controller") = True
            amount += 1
        End If
        If xbox.Checked Then
            total += 399.99
            item = "xbox"
            command = New System.Data.SqlClient.SqlCommand("UPDATE [database] SET " +
    item + "=" + "'true'" + "WHERE (name ='" + name + "')", conn)
            command.ExecuteNonQuery()
            Session("xbox") = True
            amount += 1
        End If
        If live.Checked Then
            total += 50
            item = "xboxlive"
            command = New System.Data.SqlClient.SqlCommand("UPDATE [database] SET " +
    item + "=" + "'true'" + "WHERE (name ='" + name + "')", conn)
            command.ExecuteNonQuery()
            Session("xboxlive") = True
            amount += 1
        End If
        If gow.Checked Then
            total += 59.99
            item = "gow"
            command = New System.Data.SqlClient.SqlCommand("UPDATE [database] SET " +
    item + "=" + "'true'" + "WHERE (name ='" + name + "')", conn)
            command.ExecuteNonQuery()
            Session("gow") = True
            amount += 1
        End If
        Session("amout") = amount
        Session("total") = total
        conn.Close()
        Response.Redirect("checkout.aspx?name='" & name & "'")
    End Sub
End Class
```

```
Partial Class checkout
    Inherits System.Web.UI.Page

    Protected Sub Page_Load(ByVal sender As Object, ByVal e As System.EventArgs) Handles ↙
    Me.Load
        Dim amount As String = Session("amount")
        Dim total As Double = Session("total")
        Dim tot As String = total
        Dim str As String = "for a total of $" + tot
        result.Text = str
        Dim purchased As String = "You have purchased:"
        If Session("cod") = True Then
            purchased += "Call of Duty 4, "
        End If
        If Session("gta") = True Then
            purchased += "Grand Theft Auto 4, "
        End If
        If Session("h3") = True Then
            purchased += "Halo 3, "
        End If
        If Session("h3se") = True Then
            purchased += "Halo 3 Special Edition, "
        End If
        If Session("h3le") = True Then
            purchased += "Halo 3 Legendary Edition, "
        End If
        If Session("headset") = True Then
            purchased += "Headset, "
        End If
        If Session("controller") = True Then
            purchased += "Controller, "
        End If
        If Session("xbox") = True Then
            purchased += "Xbox 360, "
        End If
        If Session("xboxlive") = True Then
            purchased += "Xbox Live, "
        End If
        If Session("gow") = True Then
            purchased += "Gears of War, "
        End If
        purchased = purchased.Remove(purchased.Length() - 1, 1)
        purchased = purchased.Remove(purchased.Length() - 1, 1)


        purch.Text = purchased


    End Sub

    Protected Sub Button1_Click(ByVal sender As Object, ByVal e As System.EventArgs)     ↙
    Handles Button1.Click
        Response.Redirect("home.aspx")
    End Sub
End Class
```

```
<%@ Page Language="VB" AutoEventWireup="false" CodeFile="checkout.aspx.vb" Inherits=
    "checkout" %>

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/
    xhtml1/DTD/xhtml1-transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml" >
<head runat="server">
    <title>Untitled Page</title>
</head>
<body id="box">
    <form id="form1" runat="server">
    <div>
    <center><h1>Checkout</h1>
        <p>
             <asp:Label ID="purch" runat="server"></asp:Label></p>
        <p>
            <asp:Label ID="result" runat="server"></asp:Label>
        </p>
        <p>
            <asp:Button ID="Button1" runat="server" Text="Return to Home Page" /> &
    nbsp;</p>
    </center>

    </div>
    </form>
</body>
</html>
```

Designed for COIN 333
Grade Received- 100

The code was compiled using the Perl compiler and Windows Vista. First goal of the program was to use Perl's sort keyword and a function to sort an array the way you want to sort it. The second part we had to open a file, read it, and calculate an average of the grades.

C:\Perl\bin\perl.exe

```
Unsorted list:
2 3 4 5 6 7 8 9 10
Sorted list:
10 9 8 7 6 5 4 3 2
The average is: 67.65
The highscore was 100 by id number : 5
Press enter to continue.
```

```
#!/usr/bin/perl

#HOMEWORK 9


#first goal was to use perls sort keyword and a fuction to sort an array they way
you wantt to sort it
#question 1

@list = (2, 3, 4, 5, 6, 7, 8, 9, 10);

print "Unsorted list:\n@list \n";

@sortedlist = sort sortlist @list
;
print "Sorted list:\n@sortedlist \n";


sub sortlist{
return $b <=> $a;
}



#question 2


open (GRADES, "grades.txt") or die "Can't open grades.txt\n";


while (<GRADES>){
($id, $grade) = m/(\w+)\s+(\w+)/;


        $sum += $grade;
$howmany++;


        if ($high < $grade){

                $high = $grade;

                $student =$id;

        }


        $value = $grade;

}


$average = $sum / $howmany;


print "The average is: $average\nThe highscore was $high by id number : $student\n";

close IN;

print "Press enter to continue.";
$line = <STDIN>;
```

Ethics paper for COIN 480
Grade received- 90

This paper was about the ethics of Artificial Intelligence in today's world.

Should Computers Think?

By Kevin Rutkowski

Artificial intelligence algorithms are seen every day in our lives whether we notice it or not. The research to make a true artificially intelligent being will persist, the question is- is it right to develop machines that can think, and have their own sense of morality without any other human interaction? The problem is, once you create a being of its own intelligence can you limit that being? Can you tell it to not have its own freewill which is the basis of being intelligent in the first place? The answer is no. We have laws in our country protecting the free will of others, freedom of speech, right to bear arms, etc. Once we create a being we need to give it the same rights as any other person even if it is just a machine that we created, even if we have to give the machine the right to bear arms. I think we should continue to develop machines that can think for themselves, although the results most likely won't come about for many, many years in the future. Why? Because robots can do things man cannot. With fiber optics robots would be able to "think" and come up with solutions to problems at the speed of light therefore contribute much to our work force in doing jobs that most humans couldn't handle easily. Not only could they think faster they could also be stronger, depending on how they are built. As of right now computers can barely learn. Marvin Minsky talks about a program called "STUDENT" in his paper and how it could solve certain high school math word problems but he wonders if the program truly understood the words that it "read" - "Then dare we say that STUDENT "understands" those words? Why bother. Why fall into the trap of feeling that we must define old words like "mean" and "understand"? It's great when words help us get good ideas, but not when they confuse us. The question should be: does STUDENT avoid the "real meanings" by using tricks?" which of course it had to, if it didn't it could talk directly back to us using the words that it knew.

But why must we continue to develop artificial intelligent beings? Because in developing them we need to take a closer look at ourselves to make them in "our image" therefore making us learn more about ourselves, which we actually know very little on how our minds work. Some of the bad implications to making robots are that they will be made to harm other humans. Making robots that would fight our wars for us is happening today but scientists are starting to refuse to make such robots: "As American warfare has shifted from draftees to drones, science and the military in the United States have become inseparable. Some scientists are refusing to let their robots grow up to be killers."(Eric Baard) The bible itself doesn't say anything about artificial intelligence, but some of it can be related to artificial intelligence easily. Paul says "there are Gods many and Lords many. I want to set it forth in a plain and simple manner; but to us there is but one God--that is pertaining to us; . . . I say there are Gods many and Lords many, but to us only one, and we are to be in subjection to that one, . . . "(History of the Church, Vol.6, p.474). So to us there is but one God, since he is our creator. Does this mean that since we create the machine are we their "god"? And would they have to answer to our God? We cannot call ourselves God because to claim there is more than one God is blasphemy. There will be many advances in artificial intelligence over the next one hundred years or so, but one thing is certain that given the proper time we will see machines thinking for themselves, acting for themselves, and making decisions like a conscious, sentient being.

Works Cited

Marvin Minsky – Why People Think Computers Can't http://web.media.mit.edu/~minsky/papers/ComputersCantThink.txt

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The Modern Reader's Bible. Richard G. Moulton editor. New York: Macmillan, 1918

Ethics Paper for COIN 315
Grade Received- 90

This ethics paper was about the ethics of online gambling and how it relates to the bible.

Ethics of Online Gambling

Kevin Rutkowski

The legality and ethics of online gambling has been an issue since the beginning of the internet. In South Carolina gambling laws include not being able to gamble on the Sabbath, keeping unlawful gaming tables (any table used for gambling), and create your own lottery and sell those tickets which results in a fine of ten-thousand dollars (Gambling-Law-Us). There have also been quite a few federal laws, acts, and statutes passed to prevent illegal gambling such as Illegal Money Transmitting Business Act of 1992, Professional and Amateur Sports Protection Act of 1992, and as far back as Travel Act of 1961 (Gambling-Law-Us). The latest was On October 13, 2006, "President Bush signed the Unlawful Internet Gambling Enforcement Act (UIGEA), which restricts the acceptance of certain financial transactions pertaining to illegal online gambling, and imposes requirements on banks and other financial service providers to identify and block illegal financial transactions." (Gameattorneys) Although none of the previous laws, acts, or statutes address the issue of online gambling directly before the UIGEA the government applied them to online gambling. These laws are put in place for the protection of the individual because in my opinion because gambling is an addiction like alcohol and drugs and it can ruin families and lives because they will spend all their money to fulfill their gambling addiction, leave houses with broken families etc. They do warn you to "be responsible" and just about all of their sites and places of businesses, but not everyone takes their warnings. With that said, gambling can also be helpful to society, the lottery, which was instated in South Carolina January 2002 which proceeds are used to fund South Carolina's education to improve the schools for the children of South Carolina- "The South Carolina Lottery has contributed over 1.7 billion dollars to the South Carolina Education Lottery fund." and "Over 28% of the money collected through game ticket sales is transferred to the Education lottery fund." (Sclottery) In addition, gambling companies provide more jobs which in turn helps the economy and for those who win, a better life.

In my opinion online gambling is fine, and I agree with Leach when he writes "Casino gambling, as it is practiced in all Western democracies, has been allowed to exist only with comprehensive regulation. Internet gambling lacks such oversight. "(online.wsj) the problem lies with the people who go too far and bet their houses, money, and other belongings just so they can play another game. It's not the site that hurts people; it's only people hurting themselves. This is the extreme end of gambling and the main reason why it should be regulated. There are a few things online gambling sites can do to keep their site regulated and make a profit at the same time. The first thing they can do is set a limit, a person pays them X amount of dollars and can gamble at their site with it for that week. Once that money is spent they cannot gamble at that site until the next week. Another idea I propose to help regulate their sites are time management. These sites could set a limit to how long a person can gamble that day so they don't end up spending more time there then they need to. The player may not like this idea, but it should be an option they can choose when they set up their profile at their site. After they limit their time and money, to help budget the player they can have them set up their own accounts to set up their own limits (time and money) that they feel they can spend a week giving them options so they don't feel like they are being held down. In the federal sense, the government can have an online gambling database, where instead of having a profile at a site the player has a national profile and can limit the player at all sites (with the ideas mentioned above), not just the one they signed up at. Of course this assumes they cannot have multiple profiles. I would also like to include this profile in any legal gambling establishment not just online, so players will be limited regardless of where they are. The players may not like these ideas but they will help limit each person (and regulate every site) while the websites still makes a profit and I would propose this worldwide.

Opponents of these ideas would complain that this limits their freedom to do what they want, but they have to realize that there is nothing in the bill of rights that states that there is a right to gamble therefore any limits put on gambling should be acceptable. Opponents of gambling itself would say that the bible condemn gambling itself but warns us to stay away from things like the love of money- "For the love of money is a root of all kinds of evil. Some people, eager for money, have wandered from the faith and pierced themselves with many griefs." , "Keep your lives free from the love of money and be content with what you have, because God has said, "Never will I leave you; never will I forsake you." , "No one can serve two masters. Either he will hate the one and love the other, or he will be devoted to the one and despise the other. You cannot serve both God and Money." (1 Timothy 6:10; Hebrews 13:5; Mathew 6:24), and the only reason people gamble is to win more money. They would also say that places like casinos and other gaming places encourage people to make un-wise decisions like spending more money by feeding them free alcohol and other things that may "help" a person to make more un-wise decisions. While this may be true, I once again have to see it's the person that makes these mistakes by accepting the free alcohol and not setting a limit for themselves on what they want to spend that night. The odds of winning most gambling games are very low, that's why it is a multibillion dollar business, but people still play anyway. Why do they gamble? In my opinion they gamble because they like the risk taking, or think they will win a huge jack pot and fix their life they aren't happy with now, but whatever the reason is they spend money doing it.

With all that said, gambling exists today, and most likely will exist well on into the future. The only thing gamblers need to learn is how to regulate their funds and time spent. By instituting some of the ideas I mention above we can help everyone realize their limits and therefore be not necessarily better gamblers, but safer and regulated.

Works Cited

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http://www.gameattorneys.com/OnlineGambling.php

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Presentation for COIN 315

I had to do a presentation for our project 6. In project 6 we had to write a c++ program that let the user input a phrase and then our program would search through a dictionary and output all anagrams of that phrase using recursive backtracking.

Project 6

COIN 315

Kevin Rutkowski

Algorithm (over view)

- Remove non alphabet characters from command line parameter “phrase”
- Use recursive backtracking method to find anagrams and keep count
- Time test the code

Algorithm (in depth)

- Store the phrase in a c++ string
- Pass the dictionary and print(true/false) to the AnagramMaker() constructor

Algorithm (in depth cont.)

- Then call the findAnagram() method passing it the c++ string as a parameter
  - Base case is when there are no letters left in the string or when you can't make any more
  - Search the dictionary for an anagram
  - Remove those letters from the c++ string
  - Recurse passing findAnagram() the rest of the phrase

Storage

- Plan to store anagrams in a vector initially then test different structures for speed
- Plan to store dictionary in a hashtable so search is order 1

Presentation for COIN 480
Grade received-100

This is my presentation on a paper involving game theory for COIN 480.

Game Theory

Kevin Rutkowski

What Is Game Theory?

- Game theory is the formal study of conflict and cooperation.
- Game theoretic concepts apply whenever the actions of several agents are interdependent.
- The concepts of game theory provide a language to formulate, structure, analyze, and understand strategic scenarios.

History and Impact of Game Theory

- The earliest example of a formal game-theoretic analysis is the study of a duopoly by Antoine Cournot in 1838.
- Game theory was established as a field in its own right after the 1944 publication of the monumental volume *Theory of Games and Economic Behavior by von Neumann and the economist Oskar Morgenstern*

Game Theory and Information Systems

- As a mathematical tool for the decision-maker the strength of game theory is the methodology it provides for structuring and analyzing problems of strategic choice.

Definitions of games

- Games can be described formally at various levels of detail
  - It typically involves several *players; a game with only one player is usually* called a *decision problem*
  - A *coalitional (or cooperative)* game

Cooperative vs Noncooperative

- Cooperative game theory investigates such coalitional games with respect to the relative amounts of power held by various players, or how a successful coalition should divide its proceeds.
- In contrast, noncooperative game theory is concerned with the analysis of strategic choices

Strategic and extensive form games

- The strategic form (also called normal form) is the basic type of game studied in noncooperative game theory
- The extensive form, also called a game tree, is more detailed than the strategic form of a game

Nash equilibrium

- A Nash equilibrium recommends a strategy to each player that the player cannot improve upon *unilaterally*, that is, given that the other players follow the recommendation. Since the other players are also rational, it is reasonable for each player to expect his opponents to follow the recommendation as well

Mixed strategies

- A game in strategic form does not always have a Nash equilibrium in which each player deterministically chooses one of his strategies
- players may instead randomly select from among these pure strategies with certain probabilities.
- Randomizing one’s own choice in this way is called a mixed strategy.

Zero-sum games and computation

- The extreme case of players with fully opposed interests is embodied in the class of twoplayer zero-sum (or constant-sum) games.